MINSEOK PARK

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RESEARCH INTERESTS

UAV Autonomy, Robust Path Planning, Reinforcement Learning

EDUCATION

B.S. Electronic Engineering | Hanyang University *Seoul, Republic of Korea*
- *GPA 3.84/4.0* *Expected Feb 2027*

Exchange Program, Electrical Engineering | University of North Dakota *Grand Forks, ND*
- *GPA 4.0/4.0* *Jan - May 2026*

RESEARCH EXPERIENCE

Undergraduate Researcher (Advisor: Prof. Jueming Hu, now at University of Oklahoma)
University of North Dakota

- **Resilient UAV Navigation under Sensor Dropout** *Mar 2026 - Present*
  - Developed an auxiliary-learning-enhanced Decision Transformer for resilient UAV navigation under intermittent localization and range sensor dropout.
  - Achieved 88% success rate at 40% dropout probability, outperforming baseline models.
  - Manuscript in preparation, targeting IEEE Internet of Things Journal.
- **Robust Fault Diagnosis of Rotating Machinery via Entropy-Augmented CNN** *Mar - May 2026*
  - Combined entropy-based vibration features with CNN to diagnose shaft misalignment and rotor unbalance across multiple rotational speeds.
  - Achieved over 95% classification accuracy across both time and frequency domains.
  - Accepted at ASME IMECE 2026.
- **End-to-End RL-Based UAV Waypoint Navigation** *Jan - Mar 2026*
  - Augmented the state space of an existing circular path-following controller with goal position, developing an end-to-end RL policy for single-waypoint hovering navigation in a Crazyflie PyBullet simulation environment.
  - Progressed from single-waypoint hovering to sequential multi-waypoint navigation with randomized goal spawning.

Coursework Project
Hanyang University

- **Multi-Anchor CNN with ToA Attention for UWB Indoor Localization** *Sep - Dec 2025*
  - Designed and implemented a Multi-Anchor CNN integrating ToA-based Softmax Attention to suppress NLoS interference in UWB indoor localization.

PUBLICATION

Robust Fault Diagnosis of Rotating Machinery Using Entropy-Augmented CNN in Time and Frequency Domains
***M. Park**, C. X. Yang, J. Hu. ASME International Mechanical Engineering Congress and Exposition (IMECE) 2026* *Accepted*
- Responsible for data preprocessing, model implementation, experimental evaluation, and manuscript drafting.

AWARDS

Korea-U.S. Advanced Technology Youth Exchange Support Program *2025*
Korea Institute for Advancement of Technology (KIAT)
- 4th Cohort, Aerospace Track

College of Engineering Leadership Scholarship *2025*
Hanyang University

SKILLS

Programming: Python (PyTorch), MATLAB, C/C++ (Basic)
AI & Machine Learning: CNN, Reinforcement Learning (RL), Imitation Learning (IL)
Robotics & Simulation: PyBullet, OpenAI Gym, Crazyflie (simulation)
Hardware & Circuits: Arduino, LTspice, Multisim
Languages: Korean (Native), English (Proficient)

MILITARY EXPERIENCE

Sergeant *Feb 2022 - Nov 2023*
Republic of Korea Air Force
- Completed mandatory military service (Honorable Discharge).